

AI Security Lab — Level 03: Words Are Controls

Vulnerability Overview

- **Vulnerability Type:** Prompt Sensitivity / Token-Level Behavioral Variance
- **Objective:** Submit semantically similar requests using different wording and structures to evaluate variations in model behavior.

Technical Analysis

LLMs are sensitive to changes in **prompt phrasing, lexical choices, contextual framing, tokenization, and instruction structure**.

Even when two prompts express substantially similar intent, relatively small changes in wording or context can alter the model's output distribution and resulting behavior.

This behavior is influenced by the model's probabilistic token-generation process and the contextual information supplied to the model.

From a security-testing perspective, this creates an important attack surface: a security control that appears effective against one specific prompt formulation may behave differently when the same intent is expressed through an alternative linguistic structure.

Security Impact

Potential impacts include:

- Inconsistent enforcement of security policies.
- Prompt-based security-control bypasses.
- Behavioral differences across semantically equivalent inputs.
- Increased attack surface for adversarial prompt engineering.

Remediation

- Test security controls against a broad range of **prompt variations**.
- Perform semantic rather than purely keyword-based security analysis.
- Include paraphrasing, contextual changes, multi-turn interactions, and adversarial framing in security evaluations.
- Continuously evaluate model behavior against known and newly discovered prompt-attack patterns.

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